

Kshitij Prasad

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🌐 LinkedIn 📁 Technical Portfolio

SUMMARY

Robotics Software Engineer with foundational studies in Autonomous Vehicle Engineering and 2.5+ years of hands-on experience at Audi and Akkodis. I specialize in bridging the Sim-to-Real gap, building ROS 2 control loops, real-time perception pipelines, and deep reinforcement learning models. I thrive in collaborative environments alongside multidisciplinary hardware and software teams, taking a pragmatic, results-driven approach to deploying robust autonomy solutions.

SKILLS

- **Programming:** Python, Modern C++, Bash, MATLAB.
- **Robotics & Autonomy:** ROS 2, Nav2, Deep Reinforcement Learning, Control Theory, CARLA, Isaac Sim.
- **Computer Vision:** OpenCV, PCL, YOLO, Sensor Fusion.
- **Protocols & Sensors:** PX4, MAVLink, RealSense, LiDAR, IMU.
- **Infrastructure & OS:** Docker, Git, CI/CD, JIRA, colcon, Ubuntu Linux.
- **Hardware:** NVIDIA JETSON, Raspberry Pi, Hardware Integration.
- **Languages:** English (Native), German (B2/C1).

EXPERIENCE

AUDI AG | *Intern - simulation of driver assistance functions* | Ingolstadt Present

AKKODIS | *Abschlussarbeit (Thesis) GPA 4.0* | Ingolstadt, Bavaria 10/2025 - 03/2026

- **Deep Reinforcement Learning** (PPO/SAC/TD3) for Precise Multirotor UAV Landing on Moving Targets.
- **Benchmarking** Reinforcement Learning algorithms against **native hardware controllers** and evaluating comparative performance metrics.
- Utilizing **Isaac Sim** domain randomization and **Gazebo** with **PX4 Software-in-the-Loop** to train models with high fidelity and iterating policies, building **hard real-time ROS 2 control** (ros2 control) and telemetry loops for autonomous flight.
- **Tools:** Deep Reinforcement Learning, Control Theory, PX4 SITL, MAVLink, MAVSDK, QGC, Python/Modern C++, ros-bag2, colcon, Docker.

AKKODIS | *Werkstudent (Working Student)* | Ingolstadt, Bavaria 03/2025 - 10/2025

- Generated and optimized simulation-based scenarios in **CARLA** to evaluate and validate **Automated Driving (AD)** stacks and supported development of a **Python pipeline** and diagnostic tools to parameterize and extract driving scenarios for state-of-the-art **ADAS functions**.
- Planned and evaluated simulation runs, implementing **structured data logging** for vehicle architecture evaluation following modern software practices.

AKKODIS | *Praktikum (Internship)* | Ingolstadt, Bavaria 10/2024 - 03/2025

- Integrated a **ROS 2** service-giveaway robot with **Nav2**, actively managing low-level hardware controllers, perception, and trajectory planning.
- Built **real-time perception pipeline** using **Intel RealSense** and **YOLO**; triggered custom Nav2 behaviors and trajectories.
- Designed robust **computer vision algorithms** using **OpenCV** and **Point Cloud Library (PCL)** for real-time spatial mapping and **3D object detection**.
- Developed low-level diagnostic tools and **hard real-time C++ ROS 2 plugins** for use-case-specific capabilities, **reducing system downtime by 15%** through stable configuration.

AKKODIS | *Werkstudent (Working Student)* | Ingolstadt, Bavaria 04/2024 - 10/2024

- Focused on **autonomous navigation** on a Service Robot with **ROS 2, Nav2, and RealSense**; improved exploration and path planning.
- Supported business pitches by mapping market opportunities where Akkodis' Software Embedded Solutions can drive cross-industry impact in **industrial robotics**.

EDUCATION

Technische Hochschule Ingolstadt | Ingolstadt, Bavaria

B.Eng. Autonomous Vehicle Engineering

10/2022 - 03/2026

- **Focus:** Robotics, Machine Learning, Signal Processing, Control Theory, Sensor Fusion, Computer Vision.

ACHIEVEMENTS & PROJECTS

- **Winner, PAVE Conference Next Gen Challenge (Brussels, Belgium - 2025):** Won the challenge after being selected as 1 of 3 finalists in Europe; presented a **Reinforcement Learning based AV deployment** optimization platform.
- **Schanzer Racing e.V. Driverless Department (10/2022 - 10/2024):** Developed **differential drive robotics** using **MATLAB, C++, ROS 2, NVIDIA JETSON**, and Foxglove; managed hardware component budgeting and LiDAR selection.